

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

A Data-Driven Overview | 1750 – 2024

Data: Global Carbon Budget 2025 • NASA • Our World in Data

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

Mauna Loa Observatory, February 2026

429 ppm

Atmospheric CO₂ (Feb 2026)

+50% above pre-industrial baseline (~280 ppm)

Acceleration of Atmospheric CO₂

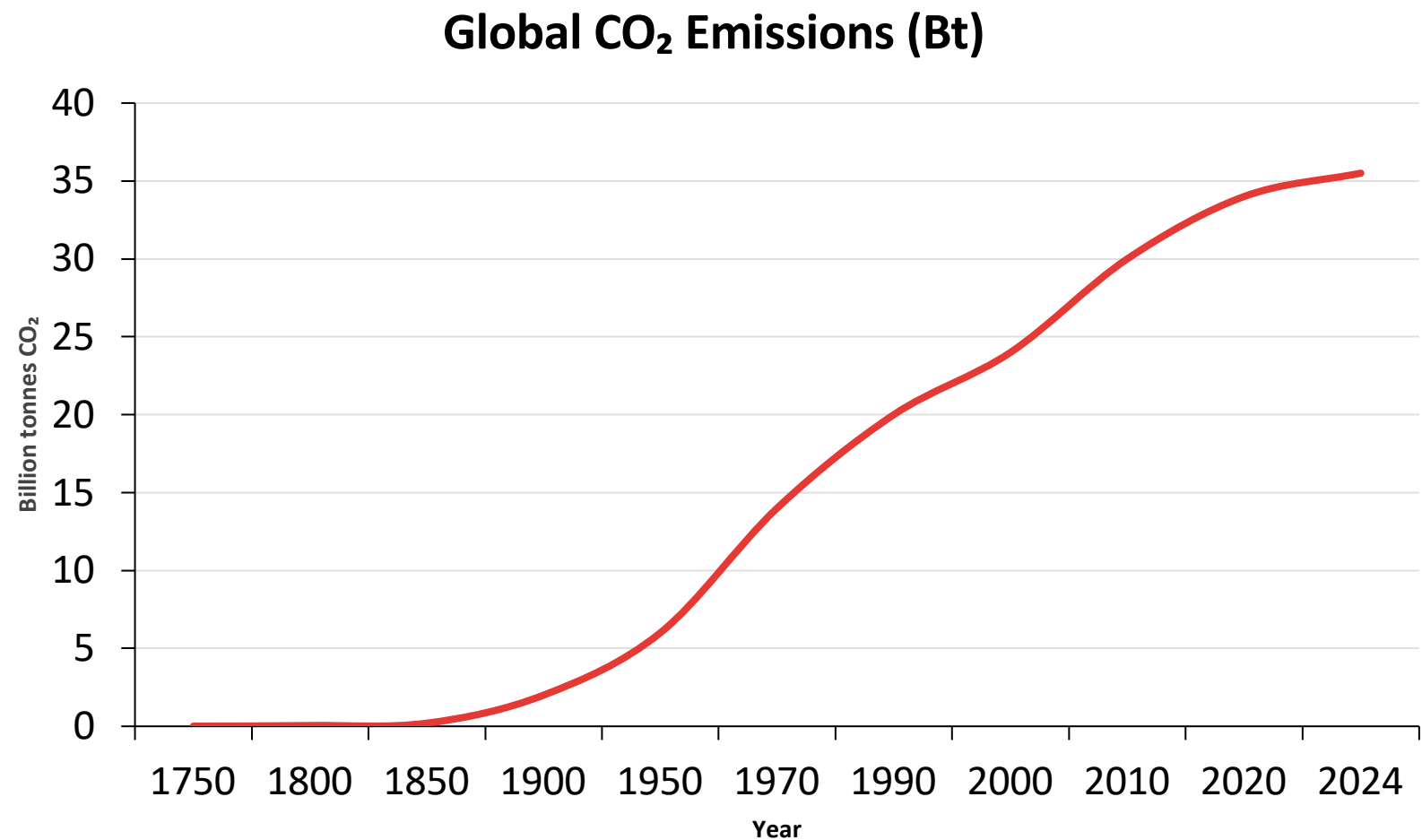
- Pre-industrial era: ~280 ppm — stable for millennia
- 1958: Mauna Loa measurements begin at ~315 ppm
- 2024: Exceeds 420 ppm — 50% above baseline
- **Feb 2026: 429 ppm recorded**

The current rate of increase is at least 1.5× greater than the natural increase observed at the end of the last ice age — a change that took thousands of years then but has occurred in ~200 years now.

Ground-based and satellite measurements confirm the sharp acceleration, linked primarily to the burning of coal, oil, and natural gas.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

1750 – 2024 trajectory with key inflection points



Key Inflection Points

1950 — 6 Bt

Post-WWII industrial expansion

1990 — 20 Bt

Nearly 4× increase in 40 years

2024 — 35+ Bt

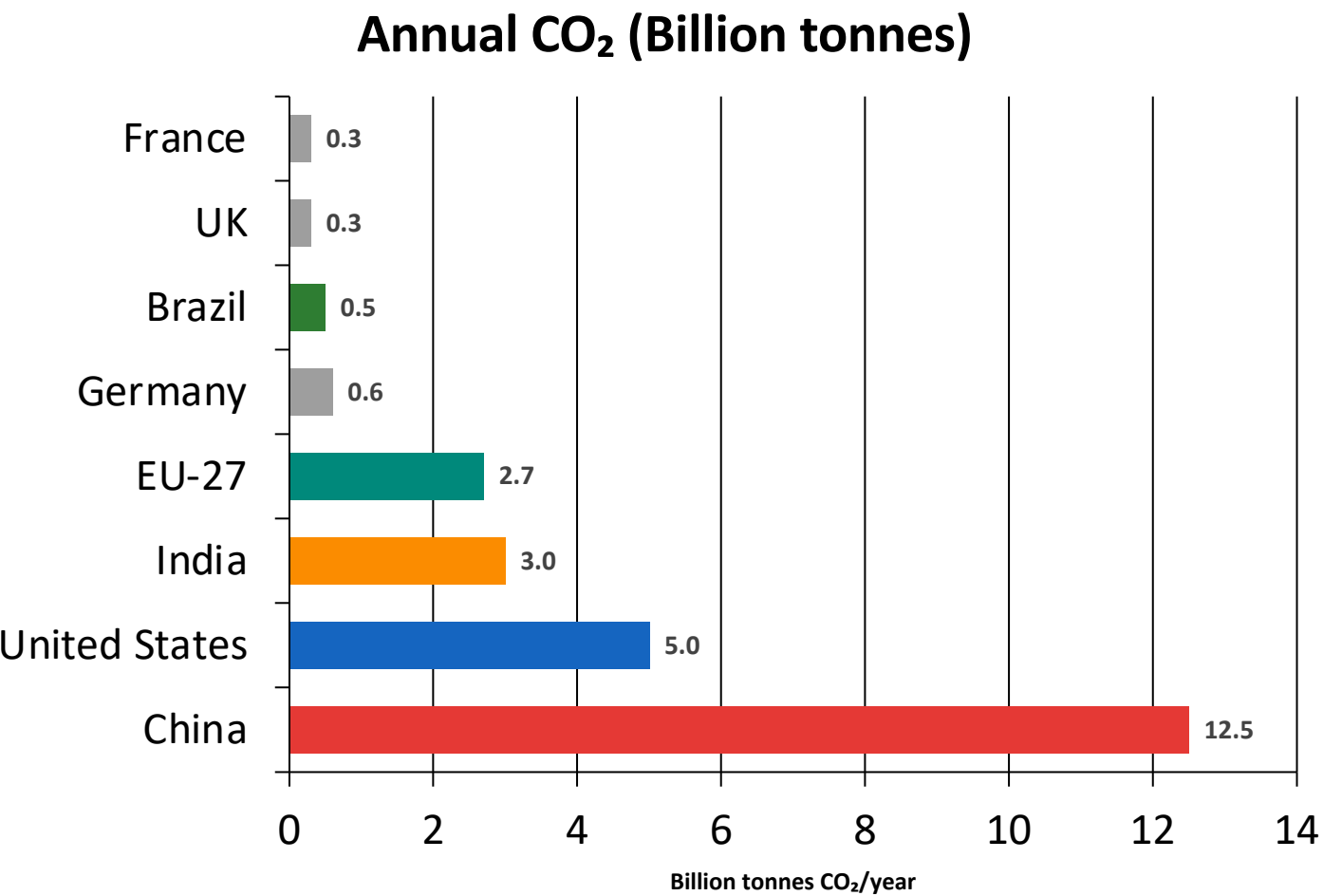
Growth slowed but not yet peaked

Before the Industrial Revolution, CO₂ emissions were negligible. Growth remained slow until the mid-20th century.

See also: figure_2.jpg — Annual CO₂ emissions by country

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Country-level annual CO₂ emissions comparison



See also: figure_3.jpg — Annual CO₂ emissions by country

Share of Global Emissions & Recent Trend

Country / Region	Share	Trend
China	~27%	Slight increase
United States	~14%	Declining
India	~8%	Rising
EU-27	~7%	Declining
Germany	~1.7%	Declining
Brazil	~1.4%	Roughly stable
United Kingdom	~0.9%	Declining
France	~0.8%	Declining

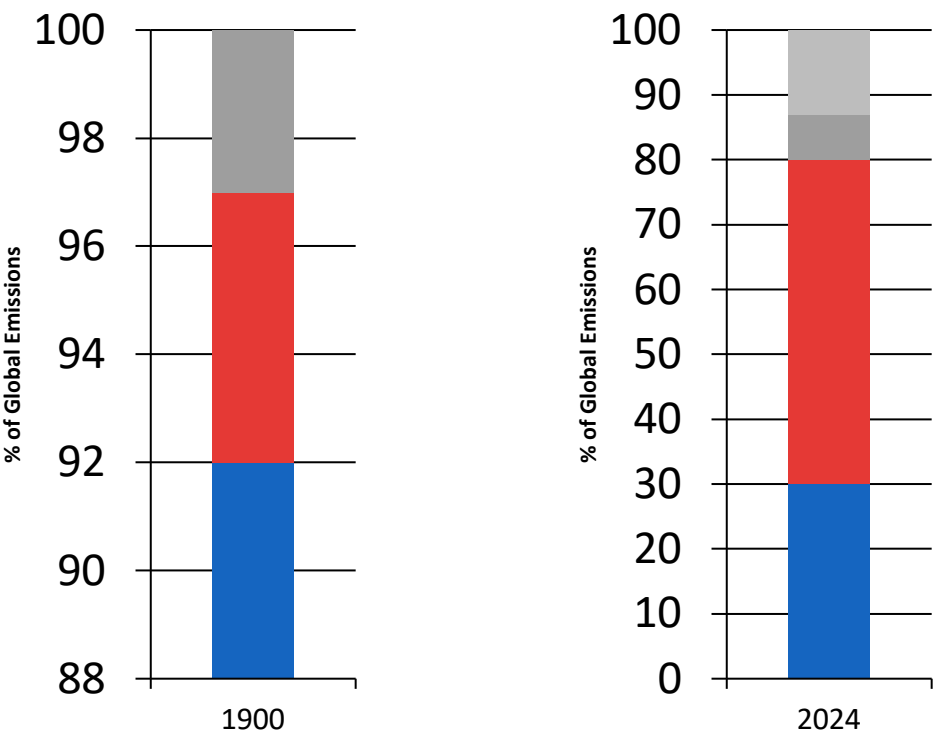
China’s steep rise post-2000 contrasts with the US decline from its ~6 Bt peak around 2005.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

Dramatic shift in regional emission shares, 1900 vs. 2024

1900: Europe + US Dominate 2024: Asia Dominates

The Geographic Center of Emissions Has Shifted



1900:

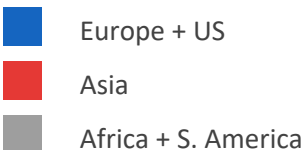
Europe and the US accounted for >90% of global emissions. By 1950, they still represented >85%.

2024:

Asia — home to nearly 60% of the world's population — produces roughly half of all emissions.

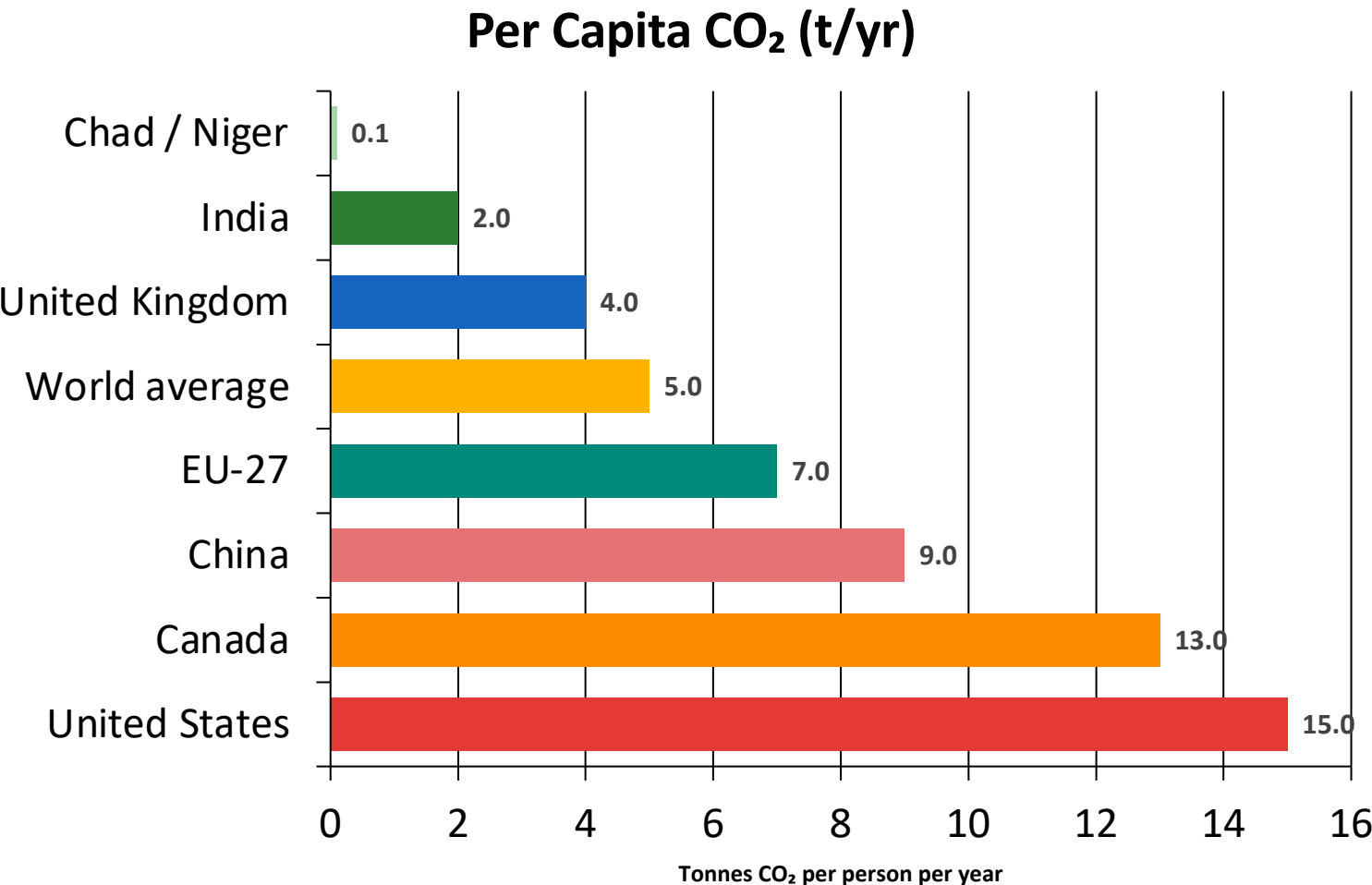
US + Europe combined: below one-third.

Africa and South America each contribute only 3–4%, comparable to international aviation and shipping.



Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Per capita CO₂ emissions for selected countries (tonnes/year)



150× DISPARITY

The average American emits in under two days what the average person in Chad or Niger emits in an entire year.

Key Observations

- US & Canada: ~3× the global average
- China: approaching EU-27 levels
- India: below world average despite being the 3rd-largest total emitter
- Sub-Saharan Africa: negligible per capita contribution

See also: figure_1.jpg — Per capita CO₂ emissions over time

Source: Global Carbon Budget 2025; NASA; Our World in Data

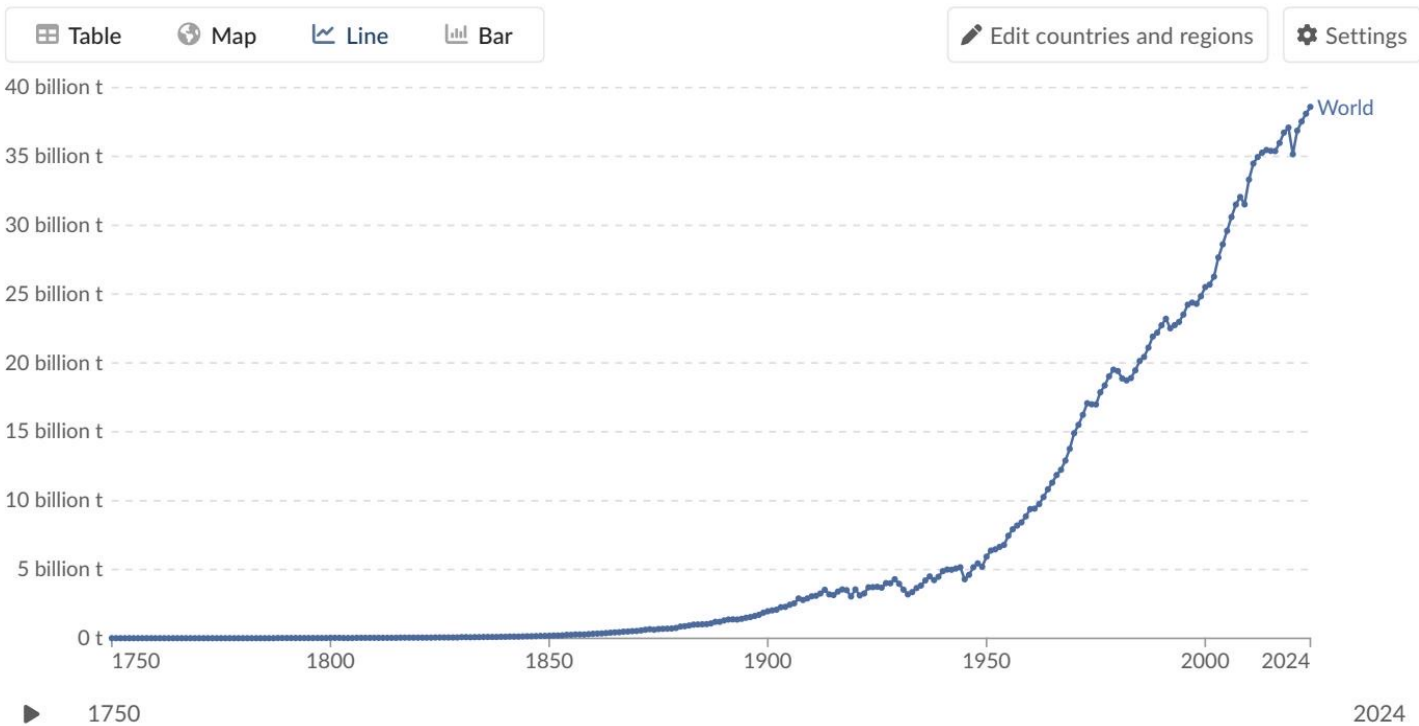
2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual percentage change in CO₂ emissions, 2024

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



Data source: Global Carbon Budget (2025) - [Learn more about this data](#)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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Regional Patterns in 2024

Declining (Blue)

United States: -1% to -5%

Much of Europe: -1% to -5%

Increasing (Orange / Red)

Russia: +2% to +10%

Parts of Southeast Asia: +2% to +10%

Several African nations: +2% to +10%+

Mixed

Some South American countries saw very large percentage increases (>10%), while others saw declines.

The 2024 map shows a mixed global picture:

mature economies are trending downward, while emerging economies continue to grow.

figure_4.jpg — Annual percentage change in CO₂ emissions, 2024

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Similar prosperity, different emissions — the role of energy sources

Countries with similar living standards can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.

	France	United Kingdom	Germany	Netherlands
Nuclear Share	High (~70%)	Moderate (~15%)	Low (~5%)	Low (~3%)
Renewables Share	Moderate	Growing (wind/solar)	Growing (wind/solar)	Growing (wind/solar)
Fossil Fuel Share	Low	Moderate	Higher (coal/gas)	Higher (gas)
Per Capita CO ₂ (t/yr)	~4	~4	~8–10	~8–10

France & UK: Higher nuclear and renewable shares → lower per capita CO₂ (~4 t)

Germany & Netherlands: Higher fossil fuel share → higher per capita CO₂ (~8–10 t)

Policy and energy mix choices — not just prosperity — determine per capita outcomes.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

Synthesis of findings

1

Global emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.

2

China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast.

3

Per capita inequalities remain extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).

4

US and European emissions are declining, but historical cumulative responsibility remains large — they dominated emissions for over a century.

5

Policy and energy mix choices — not just prosperity — determine per capita outcomes. France and the UK have much lower per capita CO₂ than Germany or the Netherlands.